

Prokaryotic Molybdenum Cofactor Biosynthesis: From GTP to Mo-Molybdopterin and Optional Dinucleotide Variants

A review-style synthesis of a conserved four-stage biosynthetic module. Search date: 2026-09-01. Citations are given as PubMed IDs (PMID).

1. Executive Summary

Prokaryotic molybdenum cofactor (Moco) biosynthesis is a short, largely linear metabolic module that transforms a single molecule of GTP into a metal-loaded pyranopterin-dithiolene cofactor and, in many organisms, appends a nucleotide to tailor the cofactor to a specific class of client enzyme. The pathway is one of the most deeply conserved in biology: homologues of its terminal enzymes are present in all three domains of life, and two of them fused during eukaryotic evolution to become gephyrin, the neuronal scaffolding protein. Despite this evolutionary depth, the chemistry is remarkable — the first committed step is a radical S-adenosylmethionine (radical-SAM) reaction that rearranges the purine skeleton of GTP into a pterin, and the sulfur-insertion step borrows a ubiquitin-like protein-conjugation logic that predates and parallels the eukaryotic ubiquitin system.

This review defines the module in four obligatory-to-optional stages. **Stage 1** builds the pyranopterin ring: MoaA, a bifunctional two-[4Fe-4S]-cluster radical-SAM enzyme, converts GTP into the cyclic intermediate 3',8-cyclo-7,8-dihydro-GTP, which MoaC then rearranges into cyclic pyranopterin monophosphate (cPMP). **Stage 2** installs the dithiolene sulfurs: the E1-like enzyme MoeB adenylates the ubiquitin-fold sulfur carrier MoaD, which — after thiocarboxylation by the cysteine desulfurase IscS — cooperates with MoaE in a MoaD₂–MoaE₂ synthase to convert cPMP into molybdopterin (MPT). **Stage 3** loads the metal: a MogA- or MoaB-class G-domain protein adenylates MPT to MPT-AMP, and a MoeA-class E-domain hydrolyzes MPT-AMP while ligating molybdate to form Mo-MPT. **Stage 4** is an optional, client-

class-specific maturation: MobA appends GMP to make bis-molybdopterin guanine dinucleotide (bis-MGD) for the DMSO-reductase family, MocA appends CMP to make molybdopterin cytosine dinucleotide (MCD) for xanthine-oxidase-type enzymes, and the sulfite-oxidase family retains the unmodified Mo-MPT.

The strongest, most reproducible conclusions of this synthesis are: (i) stages 1–3 are obligatory and strictly ordered — sulfur insertion must precede metal insertion, and MPT adenylation must precede molybdotransfer; (ii) the MogA/MoaB adenylyltransferase and MoeA insertase form an ancient, universal terminal module whose fusion produced eukaryotic gephyrin and plant Cnx1; and (iii) the dinucleotide branch is an accessory step whose product is dictated by the structural family of the downstream molybdoenzyme rather than by the cofactor chemistry itself. Well-supported uncertainties remain around the identity and regulation of the prokaryotic MoaB adenylyltransferase lineage, the precise choreography of sulfur delivery, and how cofactor forms are trafficked and protected en route to client apoenzymes.

2. Definition and Biological Boundaries

2.1 What is included

The module comprises the enzymatic conversion of **GTP → cPMP → MPT → Mo-MPT (→ bis-MGD or MCD)**. Concretely, this includes:

- **MoaA** (GTP 3',8-cyclase) and **MoaC** (cPMP synthase) — pyranopterin ring construction;
- **MoeB** (sulfur-carrier adenylyltransferase), **MoaD** (ubiquitin-like sulfur carrier), and **MoaE** (MPT synthase large subunit) — dithiolene installation;
- **MogA / catalytically competent MoaB** (MPT adenylyltransferase) and **MoeA** (molybdotransferase) — metal insertion;
- **MobA** (guanylyltransferase) and **MocA** (cytidylyltransferase) — optional dinucleotide maturation.

2.2 What is excluded (adjacent processes commonly conflated with the module)

Several neighboring processes supply or consume the pathway's intermediates but are mechanistically distinct and should be treated separately:

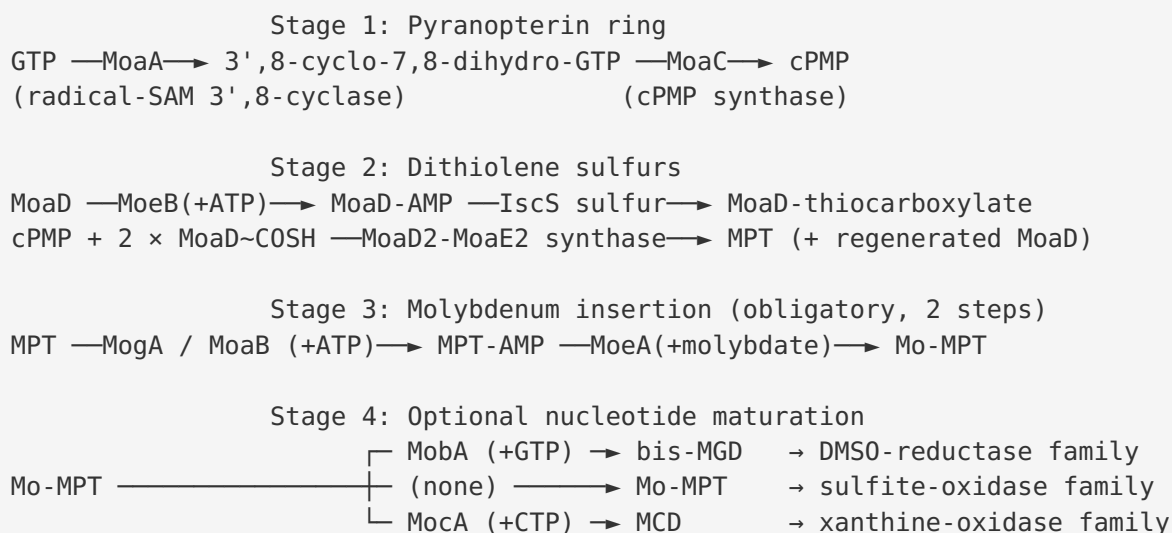
- **Upstream sulfur supply.** The cysteine desulfurase **IscS**, the rhodanese **YnjE**, and sulfur-relay proteins such as **TusA** generate the persulfide sulfur that thiocarboxylates MoaD (PMID: 19946146; PMID: 21856748; PMID: 31655739). This is the boundary between Moco biosynthesis and general cellular sulfur trafficking.
- **Iron–sulfur cluster assembly.** MoaA depends on [4Fe-4S] clusters delivered by A-type carrier proteins (ErpA/IscA), coupling cofactor output to cellular Fe-S status (PMID: 33782054; PMID: 38631442).
- **Molybdate transport** into the cytoplasm (ModABC and related systems).
- **Terminal cofactor sulfuration** of the Mo center (for xanthine-oxidase-family enzymes), performed by dedicated sulfurases downstream of the cofactor-forming module.
- **Cofactor insertion into client apoenzymes**, often via dedicated chaperones (e.g., TorD, XdhC) that bind Moco and hand it to specific apoproteins (PMID: 18522945; PMID: 17686778).
- **Mature molybdoenzyme catalysis, pathway regulation, and the eukaryotic MOCS/CNX/GPHN fusion organization and human disease**, which — while informative for evolutionary comparison — lie outside the prokaryotic module proper.

2.3 Competing definitions in the literature

Two definitional ambiguities recur. First, the **identity of the MPT adenylyltransferase** is described inconsistently: in *E. coli*, MogA performs adenylylation, but MoaB is a structural paralog of MogA whose catalytic competence has been demonstrated in archaea (e.g., *Pyrococcus furiosus*), so some lineages use a "MoaB-class" adenylyltransferase. Treating "MPT adenylylation" as a single conserved activity with alternative protein implementations resolves the confusion. Second, older literature sometimes lumps the terminal two-step Mo insertion (adenylylation + molybdotransfer) into a single "molybdenum insertase" activity — accurate for the fused eukaryotic enzymes (gephyrin, Cnx1) but potentially misleading for prokaryotes where the two activities reside on separate proteins.

3. Mechanistic Overview

The best current model is an essentially linear cascade with a single branch point at the end. The pyranopterin scaffold is built first, then decorated with two sulfurs, then charged with molybdenum, and finally — optionally — capped with a nucleotide.



Obligatory steps: Stages 1–3 are obligatory for any active cofactor. Within them, two orderings are non-negotiable: (a) sulfur must be inserted into the pterin (Stage 2) before the metal (Stage 3), and (b) MPT must be adenylylated before MoeA can ligate molybdate. **Conditional/accessory steps:** Stage 4 is conditional on the client enzyme class; its omission still yields a functional (Mo-MPT) cofactor for sulfite-oxidase-type enzymes. The requirement for MoeB-mediated MoaD reactivation is obligatory but *cyclic* — MoaD is a catalytic sulfur shuttle regenerated after every turnover.

4. Major Molecular Players and Active Assemblies

4.1 MoaA — a bifunctional two-cluster radical-SAM cyclase

MoaA initiates the pathway with chemically the most demanding step: rearranging the purine of GTP so that guanine C8 is inserted into the nascent pterin ring. Crystallographic and spectroscopic work establishes that **each MoaA monomer carries two distinct [4Fe-4S] clusters**. The N-terminal cluster is bound by the canonical CX₃CX₂C radical-SAM motif and "is

involved in the reductive cleavage of SAM and generates a 5'-deoxyadenosyl radical (5'-dA•)," while the unique C-terminal cluster is "presumably involved in substrate binding and/or activation," coordinating 5'-GTP through the guanine N1/N2 atoms (PMID: 16632608). The MoaA·5'-GTP structure visualizes the L-Met and 5'-dA SAM-cleavage products poised to abstract a hydrogen from the substrate, and — critically — shows that "the tightly anchored triphosphate moiety prevents the escape of radical intermediates," confining the radical chemistry to a productive intramolecular rearrangement (PMID: 16632608). Because MoaA "binds two [4Fe-4S] clusters per monomer" and these are delivered by A-type carriers, Moco output is mechanistically coupled to cellular iron and Fe-S status (PMID: 38631442; PMID: 33782054).

The product of MoaA is not cPMP directly but the cyclic nucleotide **3',8-cyclo-7,8-dihydro-GTP (3',8-cH₂GTP)**. Mechanistic studies from the Yokoyama laboratory show that "the characteristic pyranopterin ring is constructed by a complex rearrangement of guanosine 5'-triphosphate (GTP) into cyclic pyranopterin monophosphate (cPMP) through the action of two enzymes, MoaA and MoaC," and that "MoaC catalyzes the majority of the transformation and produces cPMP from a unique cyclic nucleotide, 3',8-cyclo-7,8-dihydro-GTP (3',8-cH₂GTP)" (PMID: 26575208). Using an uncleavable substrate analogue (3',8-cH₂GMP[CH₂]PP), they further demonstrated that the early stage of MoaC catalysis proceeds without cyclic-phosphate formation, refining the intermediate sequence. This division of labor — MoaA as the radical cyclase, MoaC as the rearrangement engine — reframes a step long attributed largely to MoaA.

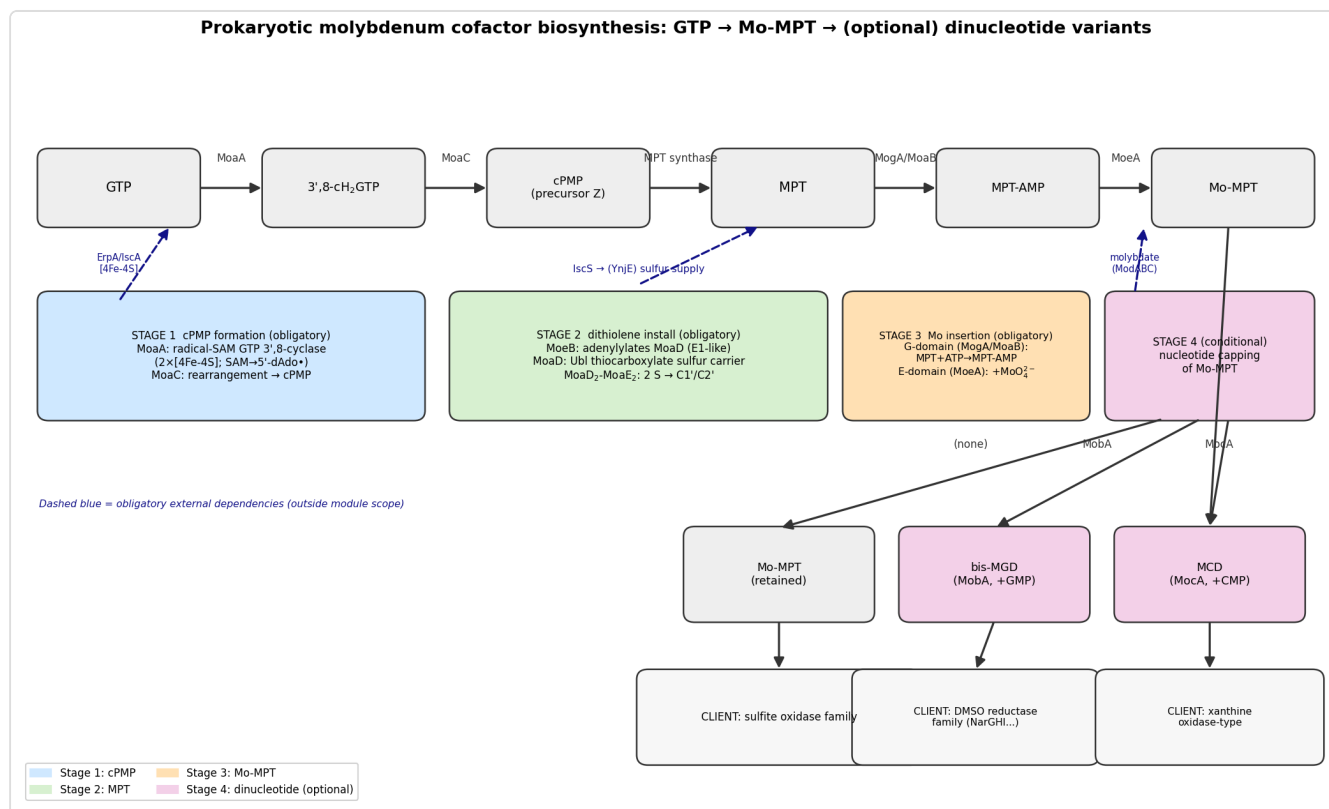


Figure 1. Schematic of the prokaryotic molybdenum cofactor biosynthesis module. GTP is cyclized by the radical-SAM enzyme MoaA and rearranged by MoaC to cyclic pyranopterin monophosphate (cPMP); molybdopterin synthase (MoaD₂-MoaE₂, reactivated by MoeB using IscS-derived sulfur) inserts two dithiolene sulfurs to form MPT; a MogA/MoaB-class G-domain adenylates MPT and a MoeA-class E-domain inserts molybdate to give Mo-MPT; MoaB or MoaA optionally append GMP or CMP to yield bis-MGD or MCD for specific client-enzyme families.

4.2 The molybdopterin synthase assembly — MoaD, MoaE, and MoeB

MPT synthase is a **heterotetramer of two large MoaE subunits and two small MoaD subunits**, with the MoaD proteins docked at opposite ends of a central MoaE dimer. It converts the sulfur- and metal-free precursor cPMP (precursor Z) into MPT: "the conversion of the sulfur- and metal-free precursor Z to MPT by MPT synthase involves the transfer of sulfur atoms from a C-terminal MoaD thiocarboxylate to the C-1' and C-2' positions of precursor Z," with the first dithiolene sulfur added at C2' (PMID: 18092812). The reaction is strictly sulfur-source-dependent: "only the thiocarboxylated MPT synthase complex was found to be able to convert precursor Z in vitro to MPT" (PMID: 11459846). Because two sulfurs are installed but each MoaD delivers one, the two-MoaD stoichiometry of the tetramer is functionally meaningful.

MoaD is a **ubiquitin-fold protein with a conserved C-terminal Gly-Gly motif**. It cycles between two heterotetrameric complexes: "MoaD cycles between two different heterotetrameric complexes, one with MoaE to form MPT synthase and the other with MoeB, a protein similar to E1 in the ubiquitin pathway, to regenerate its transferrable sulfur" (PMID: 17223713). MoeB activates the MoaD C-terminus as an acyl-adenylate (MoaD-AMP); the terminal glycine (Gly81 in *E. coli*) is essential for this MoaD-AMP formation and for downstream sulfur transfer. The MoaD-AMP is then converted to the thiocarboxylate by an external persulfide donor. This is the same chemical logic — C-terminal adenylation followed by thiol conjugation — used in ubiquitin and ThiS activation, making Moco biosynthesis a canonical example of ancestral, prokaryotic ubiquitin-like protein chemistry.

4.3 The terminal Mo-insertion module — G-domain adenylyltransferase + MoeA E-domain

Metal insertion is a **two-step reaction**. A **MogA-class (protein-G/G-domain) protein first adenylylates MPT to MPT-AMP**, and a **MoeA-class (E-domain) protein then hydrolyzes MPT-AMP and ligates molybdate to form Mo-MPT**. Structural work on *E. coli* MoaB shows it is a 3_2 -symmetric hexamer whose fold matches MogA and the G-domains of rat/human gephyrin and *Arabidopsis* Cnx1: "the overall fold of the monomer is similar to those of the MogA protein of *E. coli*, the G-domains of rat and human gephyrin and the G-domains of Cnx1 protein from *A. thaliana*, all of which are involved in the insertion of an unknown molybdenum species into molybdopterin to form the molybdenum cofactor" (PMID: 15159566). This structural identity is the basis for treating MoaB as a catalytically competent adenylyltransferase in the lineages (e.g., archaeal *Pyrococcus*) that lack a separate MogA. In eukaryotes the two activities are fused: "the final step of Moco biosynthesis, i.e. transfer and insertion of Mo into MPT, is catalyzed by the two-domain proteins Cnx1 in plants and gephyrin in mammals" (PMID: 11554796).

The molybdate-insertion chemistry itself has been dissected in the plant MoeA homolog Cnx1E. The E-domain catalyzes both reactions: "molybdate insertion and MPT-AMP hydrolysis are catalyzed by the Mo-insertase E-domain. Earlier work reported a highly conserved aspartate residue to be essential for Mo-insertase functionality" (PMID: 31860061). Substituting Cnx1E Asp274 with Glu (D274E) arrests MPT-AMP hydrolysis and causes accumulation of both MPT-AMP and molybdate; the mutant structure shows disorder of residues 269–274, attributed to the inability of Glu274 to coordinate an octahedral Mg^{2+} -water complex. High-resolution datasets further reveal "two molybdate-binding sites within the active site" whose occupancy is

tied to a distinctive backbone conformation proposed to govern molybdate selectivity ([PMID: 29717023](#)). Because Cnx1E is the direct homolog of bacterial MoeA, these mechanistic details transfer to the prokaryotic step with reasonable confidence.

4.4 The dinucleotide transferases — MobA and MocA

The optional cap is added by **paralogous, strictly nucleotide-specific transferases**. "The molybdenum cofactor is modified by the addition of GMP or CMP to the C4' phosphate of molybdopterin forming the molybdopterin guanine dinucleotide or molybdopterin cytosine dinucleotide cofactor, respectively" — via GTP:MPT guanylyltransferase **MobA** or CTP:MPT cytidylyltransferase **MocA** ([PMID: 21081498](#)). MobA and MocA share only ~22% identity yet are so nucleotide-selective that "the exchange of five amino acids was enough to obtain activity with both GTP and CTP in either MocA or MobA"; the N-terminal domain encodes nucleotide specificity while the C-terminal domain determines which client enzyme the transferase serves ([PMID: 21081498](#); [PMID: 19542235](#)).

Enzyme	Reaction	Product	Client molybdoenzyme family	Reference
(none)	—	Mo-MPT	Sulfite oxidase (SUOX) family	PMID: 18535145
MobA	Mo-MPT + GTP → bis-MGD	bis-MGD	DMSO reductase family (incl. NarGHI nitrate reductase)	PMID: 25404027
MocA	Mo-MPT + CTP → MCD	MCD	Xanthine dehydrogenase / aldehyde oxidoreductase (XdhABC, YagTSR)	PMID: 19542235

5. Evolutionary and Cell-Biological Variation

5.1 A deeply conserved, evolutionarily plastic core

The terminal insertase MoeA is essential for Moco biosynthesis and exists in all domains of life — one of the pathway's oldest components. Phylogenetic analysis reveals that "in Archaea we identified an ancestral duplication where one of the paralogs might bind tungsten instead of molybdenum" ([PMID: 39091723](#)), marking an ancient Mo/W-binding divergence at the metal-insertion step. This metal flexibility is not merely ancestral: the archaeon *Pyrobaculum*

aerophilum makes an active tungsten nitrate reductase in which "W is coordinated by a bis-molybdopterin guanine dinucleotide cofactor" (PMID: 20863064), demonstrating that the same pyranopterins and dinucleotide machinery can carry either metal.

5.2 From prokaryotic module to eukaryotic fusion proteins

The most striking evolutionary elaboration is the **fusion of MogA and MoeA in the eukaryotic lineage**: "MoeA was obtained from Bacteria by early eukaryotes, MogA fused with MoeA in the opisthokont ancestors, and it finally gained roles in anchoring inhibitory neurotransmitters" (PMID: 39091723). The resulting two-domain gephyrin/Cnx1 architecture places a MoeA-like E-domain and a MogA/MoaB-like G-domain on one polypeptide. The plant enzyme Cnx1 encodes exactly this two-domain fusion — "the N-terminal domain is homologous to the *E. coli* Moco protein MoeA, the C-terminal domain is homologous to the *E. coli* Moco proteins MoaB and MogA" — and functionally complements an *E. coli* mogA mutant (PMID: 8528286). Gephyrin later gained a moonlighting role clustering glycine and GABA receptors at inhibitory synapses (PMID: 11554796). A further example of repurposing: in Actinobacteria a MoeA copy (Glp) has lost enzymatic activity and instead functions in cell division (PMID: 39091723).

Best representative of the ancestral role: because the MoeA family has expanded and diversified (Mo- vs W-binding paralogs, enzymatically dead cell-division copies, fused eukaryotic insertases), the free-standing, catalytically active bacterial/archaeal MoeA — not the moonlighting eukaryotic fusion or the Actinobacterial Glp — is the best proxy for the ancestral molybdoxotransferase.

5.3 Physiological and lineage-specific variation

- **MPT synthase architecture varies.** Some bacteria (e.g., *M. tuberculosis*) encode a **fused MoaD–MoaE protein (MoaX)** that must be post-translationally cleaved by a JAMM/MPN protease to liberate the free C-terminal di-Gly of MoaD required for sulfur carriage (PMID: 29777693).
- **Sulfur-donor systems differ.** *E. coli* uses IscS as the primary donor with the rhodanese YnfE as a refining step — an arrangement paralleling the two-sulfurtransferase (desulfurase + rhodanese) system used in humans (PMID: 21856748).
- **Cofactor-form usage is client-driven, not organism-driven.** A single organism can make Mo-MPT, bis-MGD, and MCD simultaneously to serve its different molybdoenzyme families, distributing them to user enzymes via dedicated binding/chaperone proteins (PMID: 17686778).

6. Constraints, Dependencies, and Failure Modes

6.1 Ordering constraints

1. **MoaA before MoaC.** MoaC's substrate is MoaA's cyclic product (3',8-cH₂GTP); MoaC cannot act on GTP directly ([PMID: 26575208](#)).
2. **Sulfur before metal.** cPMP must be converted to the dithiolene-bearing MPT before molybdate can be chelated; the dithiolene is the metal-binding ligand. There is no route by which molybdate is inserted into cPMP.
3. **MoeB activation before sulfur transfer.** MoaD must be adenylylated and thiocarboxylated before it can supply sulfur to MoaE; only the thiocarboxylated synthase is active ([PMID: 11459846](#)).
4. **Adenylylation before molybdotransfer.** MoeA acts on MPT-AMP, not MPT; the D274E Cnx1E experiment, which traps MPT-AMP when hydrolysis is blocked, directly evidences this obligatory intermediate ([PMID: 31860061](#)).
5. **Metal before nucleotide (for the standard route).** Dinucleotide transferases act on Mo-MPT; notably, MocA can convert MPT to MCD in the absence of molybdate but only for a single turnover, with product remaining bound — underscoring that molybdate loading is normally coupled to productive dinucleotide formation ([PMID: 19542235](#)).

6.2 Dependency constraints

- **Iron/Fe-S dependency.** MoaA's requirement for two [4Fe-4S] clusters makes the entire pathway hostage to Fe-S cluster assembly and iron availability ([PMID: 38631442](#); [PMID: 33782054](#)).
- **Sulfur dependency.** In an *iscS* deletion strain MoaD sulfuration is greatly reduced and compound Z (the oxidation product of the immediate MPT precursor) accumulates to the same extent as in an MPT-synthase-deficient strain, whereas *csdA/sufS* deletions do not have this effect; "IscS, but not CsdA or SufS, interacts with MoeB and MoaD" ([PMID: 19946146](#)). Independently, "the l-cysteine desulfurase IscS was identified as the primary sulfur donor for the formation of the thiocarboxylate on the small subunit (MoaD) of MPT synthase" ([PMID: 21856748](#)).

6.3 Failure modes and their physiological readouts

- **Loss of dinucleotide maturation is client-specific.** MobA "converts MoCo to bis-molybdopterin guanine dinucleotide (bis-MGD), a form of the cofactor that is required by the dimethylsulfoxide (DMSO) reductase family of enzymes, which includes the nitrate reductase NarGHI"; a *M. tuberculosis mobA* deletion abolishes assimilatory and respiratory nitrate reductase activity and impairs persistence in guinea pigs (PMID: 25404027). By contrast, sulfite-oxidase-family enzymes are unaffected by *mobA* loss because they use bare Mo-MPT — consistent with the finding that MoCo-dependent resistance to N-hydroxylated base analogs in *E. coli* is MobA-independent and satisfied by the GMP-free (MPT) cofactor (PMID: 17349664).
- **Blocked molybdate insertion** (e.g., the conserved-Asp mutant of the MoeA/Cnx1E E-domain) traps MPT-AMP and molybdate, producing an inactive cofactor precursor (PMID: 31860061).

7. Controversies and Open Questions

Strongly supported claims. The radical-SAM identity and two-cluster architecture of MoaA; the MoaC-driven rearrangement to cPMP; the ubiquitin-like MoaD/MoeB activation cycle; the thiocarboxylate-dependent two-sulfur transfer by the MoaD₂–MoaE₂ synthase; the two-step (adenylylation + molybdotransfer) metal insertion; the conserved-Asp/Mg²⁺-water mechanism of the E-domain; and the client-class partitioning of Mo-MPT, bis-MGD, and MCD are all supported by convergent structural, biochemical, and genetic evidence.

Areas of disagreement or indirect evidence.

1. **Mechanistic transfer across organisms.** Much of the molybdate-insertion mechanism comes from the *plant* enzyme Cnx1E (PMID: 31860061; PMID: 29717023), whereas the adenylyltransferase paralogy comes from *E. coli* MoaB structure (PMID: 15159566) and the catalytically competent MoaB claim rests on archaeal (*Pyrococcus*) biochemistry. Whether every prokaryotic MoaB is a bona fide adenylyltransferase in vivo, or only a subset of the lineage, is not fully resolved.
2. **Identity of the physiological sulfur donor beyond IscS.** IscS is the primary donor, but the precise roles of YnjE, TusA, and other rhodanese/relay proteins — and how sulfur is channeled specifically to MoaD versus competing tRNA-thiolation and Fe-S pathways — remain partially defined (PMID: 21856748; PMID: 31655739).

3. **The Mo/W selectivity switch.** The archaeal MoeA duplication that may distinguish Mo from W ([PMID: 39091723](#)) and the tungsten-loaded bis-MGD nitrate reductase ([PMID: 20863064](#)) raise unresolved questions about how — and where in the pathway — metal identity is selected.

4. **Cofactor trafficking and protection.** How the labile cofactor is shielded from oxidation and delivered to the correct apoenzyme (via chaperones such as XdhC and TorD; [PMID: 17686778](#); [PMID: 18522945](#)) sits at the module's exit boundary and is incompletely mapped.

Most important open questions. (i) What is the full, ordered inventory of intermediates between MoaA product and cPMP, and how does MoaC accomplish the ring rearrangement? (ii) Which prokaryotic lineages genuinely use MoaB rather than MogA for adenylation, and what selects between them? (iii) How is metal specificity (Mo vs W) determined mechanistically at the MoeA step? (iv) How is the choice among Mo-MPT, bis-MGD, and MCD coordinated with client-enzyme demand in a single cell?

8. Limitations and Knowledge Gaps of This Review

- **Cross-organism synthesis.** This review integrates data from *E. coli*, *M. tuberculosis*, *Pseudomonas*, *Rhodobacter*, archaea, plants, and mammals. While the core module is conserved, mechanistic details established in one system (especially plant Cnx1E and eukaryotic gephyrin) are extrapolated to prokaryotes with appropriate but incomplete confidence.
- **The MoaB adenylyltransferase lineage** is the least directly characterized node; its inclusion as an "alternative implementation" of MPT adenylation rests on structural homology plus limited archaeal biochemistry rather than a broad enzymological survey.
- **Kinetics and flux control** across the module — which step is rate-limiting under physiological molybdate/iron/sulfur conditions — were not quantitatively assessed here.
- **Regulation** was deliberately excluded from scope, so the review does not capture how transcriptional/translational control (FNR, NarXL, Fur, ArcA; [PMID: 31517366](#)) tunes module output.

9. Proposed Follow-up Experiments and Actions

1. **Direct in vitro reconstitution of MoaB adenylyltransferase activity** across diverse prokaryotic lineages (bacteria vs archaea) to define which organisms genuinely use MoaB rather than MogA, and to test whether a single active-site signature predicts catalytic competence.
 2. **Time-resolved characterization of the MoaA→MoaC handoff**, using trapped or analogue intermediates to enumerate every species between 3',8-CH₂GTP and cPMP and to assign each rearrangement step to MoaA vs MoaC.
 3. **Structural/mechanistic dissection of Mo/W selectivity** at MoeA, comparing the archaeal Mo- and putative W-binding paralogs to identify the residues governing metal choice, complemented by molybdate/tungstate competition assays.
 4. **Systematic mapping of sulfur channeling** to MoaD, using genetic depletion and pulse-labeling to quantify the relative contributions of IscS, YnjE, and TusA and to test channeling versus free-diffusion models.
 5. **Cofactor-flux tracing in a single organism** expressing multiple client families, to determine how the cell partitions Mo-MPT among the "stop" (sulfite-oxidase), MoaA (bis-MGD), and MocA (MCD) fates in response to enzyme demand.
 6. **Cross-complementation matrix** of fused (MoaX, Cnx1, gephyrin) vs split (MoaD/MoaE; MogA/MoeA) architectures to quantify the functional consequences of evolutionary fusion and to identify the minimal active module.
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10. Key References

PMID	Contribution to this review
26575208	MoaA/MoaC two-enzyme unit; 3',8-CH ₂ GTP intermediate; MoaC does most of the rearrangement to cPMP
16632608	MoaA two-[4Fe-4S]-cluster architecture; GTP anchoring prevents radical escape
38631442	Fe-S dependence of Moco biosynthesis; two clusters per MoaA monomer
33782054	A-type carriers (ErpA/IscA) deliver [4Fe-4S] to MoaA
18092812	MPT synthase heterotetramer; two-sulfur transfer to C1'/C2' of precursor Z
11459846	Only thiocarboxylated MPT synthase is active
17223713	MoaD ubiquitin fold; MoeB (E1-like) reactivation cycle
29777693	Fused MoaD–MoaE (MoaX) cleaved by JAMM/MPN protease
19946146	IscS as the specific in vivo sulfur donor interacting with MoeB/MoaD
21856748	IscS primary donor; rhodanese YnjE refines sulfur transfer
31655739	TusA sulfur-relay diversity feeding Moco biosynthesis
15159566	MoaB structure homologous to MogA and gephyrin/Cnx1 G-domains
11554796	Terminal Mo insertion by two-domain Cnx1/gephyrin (G + E domains)
31860061	E-domain performs MPT-AMP hydrolysis + molybdate insertion; conserved Asp essential
29717023	Two molybdate-binding subsites in the Cnx1E active site
8528286	Cnx1 two-domain fusion = MoeA + MoaB/MogA homologs; complements mogA
39091723	MoeA universality, archaeal Mo/W duplication, MogA-MoeA fusion, Glp repurposing
21081498	MobA/MocA nucleotide specificity encoded by five N-terminal residues
19542235	MocA characterization; MCD required by XdhABC/YagTSR; single-turnover without molybdate
25404027	MobA/bis-MGD required by DMSO-reductase family; persistence phenotype in <i>M. tuberculosis</i>
20863064	Tungsten coordinated by bis-MGD in archaeal nitrate reductase

PMID	Contribution to this review
18535145	SUOX fold persistence across all domains; Mo-MPT-retaining client family
18522945	TorD chaperone connects cofactor synthesis to apoenzyme (exit boundary)
17686778	XdhC/MobA competition for Moco; cofactor targeting to user enzymes
17349664	GMP-free MoCo (MPT) sufficient for certain MoCo-dependent functions
31517366	Regulation of Moco/molybdoenzyme genes (excluded scope, context)

Prepared as a commissioned review-style synthesis. Claims are anchored to the primary and review literature cited above; extrapolations across organisms (notably from plant Cnx1E and eukaryotic gephyrin to prokaryotic MoeA/MogA) are flagged in the text and should be treated as mechanistically plausible rather than directly demonstrated in every prokaryote.

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